Team 12 · Paper 15 · RS3, DSBA-3, HSE · defence 17 October 2026

Price Impact or Trading Volume: Why Is the Amihud (2002) Measure Priced?

X. Lou, T. Shu, *Review of Financial Studies* 30(12), 2017

Replication on Binance spot (USDT pairs, 2019-01…2026-09) and upgrades: Amihud & Noh (2021), λ from 5-minute bars, EDGE, sentiment, MOEX, US after 2012

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Repository: `https://github.com/bigpurota/paper15-lou-shu`

Every statistic on the slides is a macro generated from `results/` (src/build_docs.py); dates and labels are typed by hand; base run 20261016T231216Z_3e8eb87e

Problem: the Amihud measure is priced, but for what?

Stocks that are illiquid by the Amihud measure earn a premium. It is usually read as compensation for price impact. Lou & Shu ask whether that reading is right.

$$A_{i,t} = \frac{1}{D} \sum_{d} \frac{|r_{i,d}|}{\mathrm{DVol}_{i,d}}$$

- The paper's premium: quintile H-L on A, 0.56 (t 2.36) %/month (NYSE/AMEX 1964–2012, 588 months).
- If the premium is for price impact, A serves both to *measure* liquidity and to *explain* returns. The paper argues that these are two different questions.

Literature: the measure (Amihud 2002) → validation against high-frequency benchmarks (Goyenko–Holden–Trzcinka 2009) → what is priced (Lou & Shu 2017) → critique of the decomposition (Amihud & Noh 2021) → spread from OHLC (EDGE, Ardia–Guidotti–Kroencke 2024).

Key observation: across assets, A is almost only volume

Dollar volume varies enormously across assets and $|r|$ varies little, so $A \approx A_C$.

	paper (NYSE/AMEX)	us (Binance)
corr(A, A_C)	0.899	0.956
Q75/Q25 of A_C	116.60	9.40
Q75/Q25 of $\|Ret\|$	2.18	1.75
corr(A, $\|Ret\|$)	0.489	−0.133

- Our dispersion of liquidity is an order of magnitude smaller (the volume filter and Binance's listing policy are a hypothesis for why), but the conclusion is the same: A is close to A_C.
- Difference: for coins, volume rises with volatility, so corr(A, $|Ret|$) is negative.

Source: results/20261016T231216Z_3e8eb87e/paper_vs_us.csv (T1)

Method: a “constant” measure and a residual

If A_C is priced and the part of A beyond A_C is not, the Amihud premium is a volume premium.

$$A_C = \frac{1}{D} \sum_d \frac{1}{\mathrm{DVol}_d}$$

Res.$\ln A$: the residual of a cross-sectional regression of $\ln A$ on $\ln A_C$, estimated each month.

“Volume” hypothesis
A_C is priced, the residual is not

“Price impact” hypothesis
the residual Res.A is priced

First we tested on simulations whether the pipeline tells the hypotheses apart (300 coins × 90 months; 3 of 17 criteria fail: world-0 size for Res.lnA (2), world-P Res.A sort):

- world V (volume premium): A_C significant in 100% of runs, Res.lnA in 4%;
- world P (price-impact premium): Res.lnA significant in FM in 100%, in the quintile sort only in 76%; but $\ln A_C$ is also falsely significant in 18%.
- Simulated noise is smaller than real noise: the real SE of the A_C spread is about 3.4–3.5 times the simulated one (an estimate), so power on real data is much lower.

Source: results/simulations/sim_world_summary.csv; results/upgrades/robustness/

Method: tests and data

Quintile sorts and Fama–MacBeth, with the signal from month $t-2$ and Newey–West(6) t-statistics.

measure in month $t-2$ $\rightarrow$ skip $t-1$ $\rightarrow$ return in month t

$$r_{i,t} - \hat{\beta}_i \, \mathrm{CMKT}_t = \gamma_0 + \gamma_1 X_{i,t-2} + \text{controls} + \varepsilon_{i,t}$$

Paper: NYSE/AMEX 1964–2012, 588 months, 1775 stocks per month; FF3-adjusted returns.

Us: Binance spot USDT pairs, later-delisted pairs kept (count not recorded; no delisting returns, D10); 93 months, on average 265 coins (626 in total); FM on 87 months with 214 coins. CMKT instead of FF3 (no size, no B/M).

- Data audit: 8 redenominations adjusted like splits, 2 reused tickers split into separate assets (dates match trading halts).
- Base specification fixed before the first real-data run; its config hash matches the copy in docs/preregistration/. The main run uses config 3e8eb87e, which adds the data audit (J1) and a robustness-only option (J2).

The paper's results (Table 3, Table 4)

In the paper the spreads on A and A_C are positive and significant, the spread on Res.A is not; in FM, Res.lnA is significantly negative.

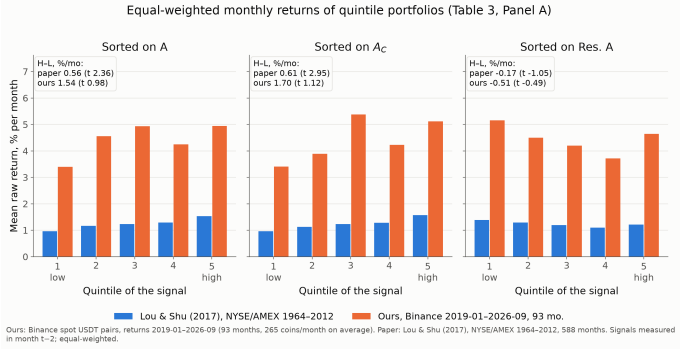
signal	quintiles Q1…Q5, %/month	H-L
A	0.96 / 1.16 / 1.23 / 1.29 / 1.53	0.56 (t 2.36)
A_C	0.96 / 1.13 / 1.23 / 1.28 / 1.57	0.61 (t 2.95)
Res.A	1.39 / 1.29 / 1.19 / 1.10 / 1.21	−0.17 (t −1.05)

- FM: $\ln A_C$ 0.183 (t 4.79); Res.lnA −0.383 (t −5.07).
- The paper's logic: the part of A beyond volume is priced with a negative sign, so the premium is not for price impact.

Source: src/paper_numbers.yaml (transcribed with pdftotext, checked against the PDF by hand; page references in the file; no automated check)

Our replication next to the paper

The sign of H-L matches (Q5 > Q1 for A and A_C, negative for Res.A), but our quintiles are not monotone and our spreads are noisy and insignificant.



H-L A_C: 1.70 (t 1.12) (paper 0.61 (t 2.95))

H-L A: 1.54 (t 0.98)

H-L Res.A: −0.510 (t −0.49)

s.d. of a coin's monthly return: 43.9%.

Without March 2021 (altcoin season) the A_C spread is 0.09 %/month (t 0.08).

Source: results/20261016T231216Z_3e8eb87e/fig_quintiles.png: the paper (Table 3, Panel A) and us on the same axes

The paper versus us

Signs agree for 25 of 28. The paper's significance repeats in 8 of its 21 significant rows, mostly Res.lnA (6 of 7; 5 at the world-0 cut-off, CI 2.04–2.96); none at HLZ.

metric	paper (NYSE/AMEX 1964–2012)	us (Binance)	match
corr(A, A_C)	0.899	0.956	yes
Q75/Q25 of A_C	116.60	9.40	no
corr(A, |Ret|)	0.489	−0.133	no
H-L on A, raw (%/mo)	0.56 (t 2.36)	1.54 (t 0.98)	sign only
H-L on A_C, raw (%/mo)	0.61 (t 2.95)	1.70 (t 1.12)	sign only
H-L on A_C, alpha (%/mo)	0.44 (t 3.20)	0.781 (t 0.62)	sign only
H-L on Res.A, raw (%/mo)	−0.17 (t −1.05)	−0.510 (t −0.49)	yes
FM ln A, model (1)	0.119 (t 3.01)	0.580 (t 1.24)	sign only
FM ln A_C, model (2)	0.183 (t 4.79)	0.760 (t 1.53)	sign only
FM Res.lnA, model (3)	−0.383 (t −5.07)	−5.09 (t −2.46)	yes
FM Res.lnA, model (4) with ln A_C	−0.248 (t −3.14)	−8.10 (t −2.37)	yes
alpha of H-L(A) on IMLA_C (%/mo)	−0.04 (t −0.77)	0.234 (t 0.50)	yes*

* Both insignificant, opposite signs. Also repeated: ln|Ret| in Table 2 (−9.05 (t −2.50)), ln A_C in the first half (1.98 (t 2.59)). World-0 cut-off 2.15–2.16 (models (3)/(4) only); below it: Res.lnA with IVOL (−6.33 (t −1.98)), on raw returns (−3.70 (t −2.11)).
Why the volume premium may be insignificant: (1) power (measured): 93 vs 588 months, MDE80 of the A_C spread 4.27 %/month; untested hypotheses: (2) another market, where volume rises with volatility; (3) no size or B/M control (CMKT, not FF3).

Source: All 28 rows: README.md, paper_vs_us.md

HF benchmark: λ from 5-minute bars, trade sign from taker volume

A correlates with λ at 0.615 (paper 0.737). In one 36-month window ln λ has a negative FM slope (−2.59 (t −4.03); size of NW(6) at this T not calibrated), unlike Table 9 model (1) (with size and B/M controls that we lack); the A_C premium is absent in this window, so this may be a window effect.

	paper (T9, T10: λ sample 1983–2012, 358 mo.)	us (Binance 2023-10…2026-09, 5-min bars)
corr(λ, A)	0.737	0.615
corr(λ, A_C)	0.747	0.547
corr(λ, Res.A)	0.149	0.294
FM ln λ, model (1)	−0.034 (t −0.67)	−2.59 (t −4.03)
FM ln λ with ln A_C, model (2)	−0.155 (t −2.51)	−4.30 (t −3.57)
FM ln A_C with ln λ, model (2)	0.18 (t 3.80)	1.06 (t 1.97)
T10: cost part, model (3)	0.015 (t 0.35)	−1.42 (t −3.92)
T10: non-cost part, model (3)	0.191 (t 3.10)	0.957 (t 1.88)

- 13 708 coin-months; sign from the aggregated taker-buy volume of 5-minute klines (no Lee–Ready). Bar λ vs trade-level λ from aggTrades: corr of ln λ 0.86 (Spearman 0.80) on 69 coin-months (2024-06, 2025-06); bar λ is 8.1 times larger in level, so only logs and ranks compare.
- One short window; NW(6) size is not calibrated at this T. Without λ, $\ln A_C$ is −0.435 (t −1.56) here (negative, insignificant). Post hoc checks: without exit months, model (2) ln λ −3.36 (t −2.75); with IVOL, model (1) −2.11 (t −2.93); with dollar volume, model (1) −3.92 (t −3.78).

Source: results/upgrades/hf_lambda/ (T9A_corr, T9B_fm, T10_fm, validation_summary); paper Tables 9–10

Liquidity or mispricing: sentiment, market illiquidity, January

Sentiment: L−S insignificant (2.57 (t 1.84)); the main FM slope 0.586 (t 1.99) passes 1.96 only marginally and not HLZ (size uncalibrated). Market illiquidity has the paper's sign, but it weakens with a market control (post hoc).

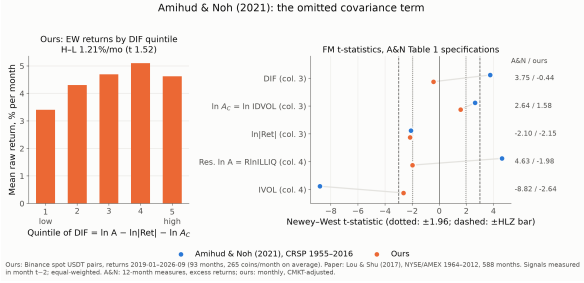
test	paper (AT_C; BW / PS)	us (A_C; Fear & Greed / AR(2) of A)
Sentiment → L−S (Table 13)	0.24 (t 1.81)	2.57 (t 1.84)
Sentiment → short leg	−0.25 (t −2.67)	−1.37 (t −1.70)
Sentiment → FM slope of ln A_C	—	0.586 (t 1.99)
Market illiquidity → L−S (Table 12)	−0.048 (t −2.38)	−4.30 (t −3.58)
same, + CMKT(t−1), post hoc	—	−3.16 (t −2.60)
January slope (Table 11 (1); paper: ln AT_C)	−0.367 (t −2.27)	January 0.699 (t 0.39), n 7
Other months	0.23 (t 6.28)	0.766 (t 1.62), n 80

- Sentiment: 0 of 4 L−S variants not labelled post hoc pass 1.96; world 0 did not calibrate the FM slope.
- Our illiquidity state, an AR(2) residual of mean *A* fitted on the full sample (look-ahead), also tracks market volatility, so we do not claim "against liquidity".
- Only 7 Januaries: for us January is a placebo. The earnings-announcement test cannot be run: coins have no earnings.

Source: results/upgrades/sentiment/paper_vs_us.csv, panel_b.csv; results/20261016T231216Z_3e8eb87e/T7_january.csv

Who is right: Lou & Shu or Amihud & Noh?

The A&N premise holds in crypto, but DIF is insignificant and our power against an A&N-sized effect is unknown: the data do not settle the dispute.



$$\ln A = \ln |Ret| + \ln A_C + \mathrm{DIF}$$

COV < 0 in 98.8% of coin-months

DIF given $\ln A_C$, $\ln |Ret|$: −1.64 (t −0.44) (monthly), 14.11 (t 1.55) (12-month, A&N's sign); A&N 0.996 (t 3.75); scales differ, so power against A&N's effect is unknown

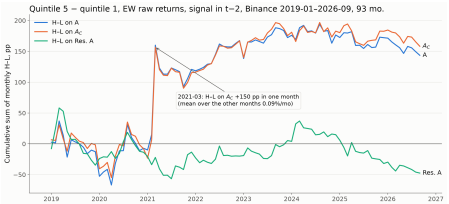
fDIF (12-month): 45.12 (t 2.14), the only main-specification result toward A&N, below HLZ (raw returns: monthly DIF 5.43 (t 2.03))

Res.lnA: −6.33 (t −1.98) with IVOL, −0.143 (t −0.05) with $\ln |Ret|$ + IVOL; R^2 of Res.lnA on $\ln |Ret|$ + DIF = 0.87. "Residual = volatility" is a hypothesis.

Source: results/upgrades/amihud_noh/; verdict rule (not pre-registered): none / AN_weak / none at minimum coins k+10 (main) / 30 / 50 (sensitivity added post hoc)

Critique

The paper's weak link is the interpretation of the residual; ours is power and one month.



1. The decomposition drops the covariance term (A&N). In our data the residual is mostly $\ln|Ret|$ (R^2 0.61); by construction it is almost entirely $\ln|Ret|$ + DIF (R^2 0.87). The paper controls IVOL (Table 4, model 5), and IVOL alone does not absorb it for us either (−6.33 (t −1.98)); the IVOL-puzzle link is a hypothesis.
2. "Volume, not liquidity" may be a false dichotomy: liquid assets trade more (Amihud–Mendelson 1986).
3. Hypothesis, untested: Lee–Ready errors could attenuate the paper's λ (the paper checks its λ against Hasbrouck's estimates). Our λ is priced negatively, so "a noisy λ hides a premium" is not supported by our data.
4. US before 2012 only: after 2012 the raw OpenAP Illiquidity L−S fell (0.324 (t 2.78) → −0.142 (t −1.47)); its FF3+UMD alpha is insignificant in both periods (0.136 (t 1.60) → 0.165 (t 1.66)) and the SMB loading fell (0.705 (t 14.93) → 0.241 (t 4.46)): consistent with lower size exposure, not tested formally. OpenAP (NYSE only, price above 5 dollars, annual rebalancing) is not the paper's design (NYSE/AMEX, monthly A at $t-2$, monthly rebalancing).

Our side: without 2021 the A_C spread is 0.432 (t 0.42). On MOEX the positive $\ln A_C$ slope repeats at 1.96 (model (2) 0.308 (t 2.87); (1) and (5) at HLZ, B5; raw returns: model (2) 0.114 (t 1.29), model (4) 0.100 (t 1.15)), but the paper's negative Res.lnA does not (−0.038 (t −0.13)).

Overfitting: could these results be overfitted?

Authors: subperiods, NASDAQ out of sample, annual measures, raw returns.

Us:

- Base specification fixed before the first run (time order: our version history); later changes J1, J2 logged. In that first run no T4 slope reached 1.96 (Res.lnA (3) −128.81 (t −1.06)); the J1 audit rules were designed after it, so the Res.lnA significance depends on a post-first-run data fix. Upgrade specifications and verdict rules were not pre-registered; post hoc variants are labelled.
- Ledger of every t: base family 73 tests, |t| > 1.96 for 16 (3.65–5.7 expected under the null; most are correlated variants of the Res.lnA hypothesis, so this is not a formal test); |t| > 3 for 0; Bonferroni/BH 0/0.
- Size in world 0 (a balanced, lighter-tailed panel): NW(6) gives 6.5%–9.5% false |t| > 1.96; empirical 5% cut-off 2.00–2.22. Simulations: 14 of 17 criteria pass (5 under a loose "share ≤ 20%" rule; 11 with 5%); failed: world P, hl_ResA: 76.0% (rule ≥ 80%); world 0, fm3_ResLnA: 9.5% (95% CI 5.8%–14.4% excludes 5%); world 0, fm4_ResLnA: 9.5% (95% CI 5.8%–14.4% excludes 5%).
- Month-block bootstrap: a CI excluding zero is stable for only 2 of 11 statistics (both Res.lnA). Placebo: shuffling the measures within month (20 permutations) gives |mean t| $\leq$ 0.32, s.d. of t 0.77–1.12, share |t| > 1.96 $\leq$ 10%; a lag-0 signal (contemporaneous, not a pure leak test) gives |t| up to 6.02.
- Survivorship: later-delisted pairs kept (count not recorded), no delisting returns. Costs: break-even 1.52% per side, from an insignificant gross spread. Non-stationarity: $\ln A_C$ 1.98 (t 2.59) in the first half, −0.280 (t −1.09) in the second.
- Out of sample: MOEX (variants not pre-registered), US after 2012 (ready-made OpenAP portfolios). Robustness, not out of sample: without 2021.

Source: results/upgrades/robustness/ (ledger_summary, bootstrap_ci, world0_calibration); results/20261016T231216Z_3e8eb87e/placebo.csv

Next steps: what we would do with three more months

The priority is power and a proper spread measurement.

1. Coin market capitalisation (supply from CoinGecko/CMC) → size control, AT/AT_C and VW portfolios.
2. Intraday EDGE on 5-minute bars and spreads from order-book snapshots.
3. λ from midquotes and over the full history, not over 36 months.
4. The A&N test on 12-month measures with a minimum number of coins fixed in advance; the residual on $\ln$ VOLUME, as in Table 6.
5. McLean–Pontiff: post-publication decay of the premium within the US sample.

MDE80 for FM $\ln A_C$: 1.39 per unit of $\ln A_C$ against the paper's slope 0.183 (7.6 times larger); per 1 s.d. 2.15 against 0.46 (4.6 times). We need more months or less noise, not more specifications.

Backup

B1. Definitions

- $A = \mathrm{mean}_d(|r_d|/\mathrm{DVol}_d)$, $A_C = \mathrm{mean}_d(1/\mathrm{DVol}_d)$; Res.lnA is the residual of $\ln A$ on $\ln A_C$, estimated each month.
- Res.$\ln A$ regression: $\ln A = a_t + b_t \ln A_C + \mathrm{Res.}\ln A$.
- Amihud & Noh: $\ln A = \ln |Ret| + \ln A_C + \mathrm{DIF}$; DIF reflects $\mathrm{cov}(|r|, 1/\mathrm{DVol})$.
- λ (Hasbrouck): 5-minute return on $\sum \mathrm{sign} \cdot \sqrt{\mathrm{DVol}}$; ours uses SVol $= \sqrt{B} - \sqrt{S}$ from taker volume per bar.
- IMLA$_C$: tercile H-L on A_C; CMKT: EW market of the universe; CMOM: tercile H-L on Ret$[-12,-2]$.

Source: src/measures.py, src/hf_lambda.py, src/factors.py

B2. Simulations: worlds V / P / 0

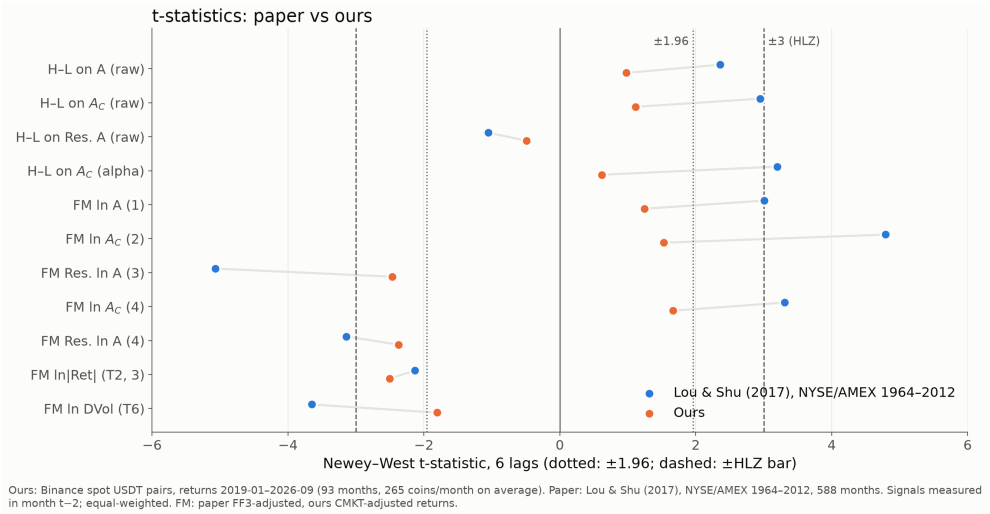
In world V A_C is priced, in world P Res.lnA; in world 0 NW(6) false positives are above 5%, significantly so only for Res.lnA (9.5%; a P0 criterion fails).

world	statistic	expectation	share significant	mean t	replications
V	fm2_lnA_C	priced: high share of t > 1.96	100.0%	7.24	50
V	hl_A_C	priced	100.0%	6.61	50
V	fm3_ResLnA	not priced: share of |t| > 1.96 near 5%	4.0%	0.06	50
V	hl_ResA	not priced	8.0%	−0.05	50
P	fm3_ResLnA	priced	100.0%	4.78	50
P	hl_ResA	priced (the sort is weaker than FM)	76.0%	2.83	50
P	fm2_lnA_C	ideally not priced	18.0%	0.01	50
0	fm2_lnA_C	false positives near 5%	8.0%	−0.03	200
0	fm3_ResLnA	false positives near 5%	9.5%	−0.14	200
0	fm3_ResLnA_nw0	same, without NW lags	5.0%	−0.12	200
0	hl_A_C	false positives near 5%	6.5%	−0.06	200

14 of 17 criteria pass; 5 of them use the loose rule “share ≤ 20%” for unpriced signals (world P falsely prices $\ln A_C$ in 18%); with 5% instead, 11 pass. Failed: world P, hl_ResA: 76.0% (rule ≥ 80%); world 0, fm3_ResLnA: 9.5% (95% CI 5.8%–14.4% excludes 5%); world 0, fm4_ResLnA: 9.5% (95% CI 5.8%–14.4% excludes 5%).

Source: results/simulations/sim_world_summary.csv, sim_checks.csv

B3. The paper's t-statistics and ours



Source: results/20261016T231216Z_3e8eb87e/fig_tstats.png (lines at 1.96 and HLZ 3)

B4. Power and robustness

- MDE80: H-L A_C 4.27 %/month against the paper's premium 0.61. FM $\ln A_C$: 1.39 per unit of $\ln A_C$ against 0.183; per 1 s.d. (each in its own s.d.) 2.15 (s.d. 1.54) against 0.46 (paper's s.d. 2.53).
- Without 2021: H-L A_C 0.432 (t 0.42), FM $\ln A_C$ 0.351 (t 0.97), Res.lnA (3) −4.93 (t −2.54).
- 31 of 45 variants keep the base verdict (same sign, same side of 1.96): 24 of them are insignificant estimates that stay insignificant, and 7 of 8 Res.lnA variants stay significant. Variants: winsorisation, raw returns, weights, subperiods, and 7 post hoc variants without the most influential month. Subperiods: $\ln A_C$ 1.98 (t 2.59) and −0.280 (t −1.09).
- Bootstrap: FM Res.lnA (3) [−8.98; −1.52], (4) [−15.29; −2.45]; FM $\ln A_C$ (4) borderline (11/20 seeds exclude zero).

Source: results/upgrades/robustness/; results/20261016T231216Z_3e8eb87e/robustness_vs_base.csv

B5. Other upgrades

- **EDGE and costs:** H-L A_C gross 1.65 (t 1.07), net 1.57 (t 1.02); daily EDGE does not measure the spread (gap weight 0.9996); break-even cost 1.52% per side, computed from an insignificant gross spread that rests on March 2021.
- **MOEX TQBR:** FM $\ln A$ 0.313 (t 3.01) (at the HLZ bar: FM (1) lnA t 3.01; FM (5) lnA_C t 3.19; FM (5) lnAT_C t 3.04; NW size on MOEX uncalibrated), $\ln A_C$ 0.308 (t 2.87), Res.lnA (4) −0.038 (t −0.13) (paper −0.248 (t −3.14)); A_C spread 1.11 (t 1.53), Res.A spread 0.462 (t 1.23) (paper −0.17 (t −1.05)), alpha 0.034 (t 0.13), SMB loading 0.906 (t 8.54); FM with $\ln$ ME keeps $\ln A_C$ (0.298 (t 2.85); ME from today's share count, a look-ahead). Post hoc: 2023 gives 65% of the total spread.
- **US (OpenAP):** Illiquidity 0.324 (t 2.78) → −0.142 (t −1.47), FF3+UMD alpha 0.136 (t 1.60) → 0.165 (t 1.66); DolVol 0.719 (t 3.98) → −0.013 (t −0.09); t of the raw difference by NW lags 0 / 3 / 6 / 12: −2.58 / −3.00 / −3.08 / −2.97.
- **Test ledger:** base tests and upgrades together, 750 unique tests, |t| > 3 for 66, 14 after Bonferroni: hf_lambda: ln λ 5, T10 cost part 2, lnA_C 1; moex: factor loading 5; sentiment: Long-Short 1. These are factor loadings, 8 rows from the one 36-month λ window (7 λ or cost-part rows, 1 lnA_C row) and one non-main market-illiquidity variant, not distinct findings.

Source: results/upgrades/*/SUMMARY.md; results/upgrades/robustness/ledger.csv

B6. Reproducibility

- `make all` runs data → tests → simulations → base run → report → upgrades → ledger → documents → manifest check; `make slides` builds this deck.
- **make all on a fresh clone** of the final code (results/repro/final_clone/): exit code 0, about 9 min; code tests 172 passed, output-manifest contract 13 passed, 3 xfailed (xfails: historical runs); data/ a directory of the clone (a copy of the cached downloads). Its outputs are the canonical results (9 run manifests); base CSVs vs the previous base run: max abs diff 0. Verdict: reproduced.
- **From-scratch clone** of an earlier version of the code with an empty data/ (every download redone): venv → data → tests → p0 → upgrades-light → upgrade-hf_lambda (3 attempts); hf_lambda needed reruns after network drops; earlier attempts exited 0 despite failed downloads (the code now fails); only the complete last one was compared. Against the current canonical outputs: of 119 result CSVs (timing columns excluded), 0 differ numerically (9 only in text; 2 agree in totals or shared rows); daily panel identical (788 692 rows). Verdict: reproduced.
- Every run writes `run_manifest.json` (sha256 of config, code and data, seed, timings); `make verify` recomputes them from the files, no git needed: config all 10 match, code all 10 match the code here (list: README). Canonical manifests failing the path check: 0; home paths remain only in history files.
- Deviations from the paper: DEVIATIONS.md (D1–D18, U1–U8, J1–J2).

Source: README.md, section “Reproducibility”; results/repro/ and results/repro/final_clone/

B7. What was not done and limitations

- **Reproducibility:** no single clean-clone run both redid the downloads and ran `make all` (`make all` of the final code used cached data; the from-scratch clone ran an earlier version of the code, and its hf_lambda step needed 3 attempts); macOS only.
- **Simulations:** the world-0 size criterion fails for Res.lnA (9.5%); 5 passes use a loose "share ≤ 20%" rule.
- **Pre-registration:** the Res.lnA significance appears only after the J1 audit, designed after the first run; verdict rules and upgrade specifications were not pre-registered; post hoc diagnostics (λ checks, MOEX same-months check, without the most influential month) are labelled.
- **Unevidenced checks:** no stored announcement sources, no saved cache comparison; the number of delisted pairs is not recorded; no delisting returns; the paper's numbers were checked against the PDF by hand, with no automated test.
- **Not done:** market caps (size control, VW by cap); the t+1 leak placebo; corr(AT_C, λ); λ from midquotes.
- **Formalities:** the slides were due 24 hours before the defence (16 October, 13:00), but the deck was first saved on 2026-10-16 19:26 (the first version of slides/deck.tex in our version history): deadline missed. Latin spelling of the names not confirmed by the members; no timed rehearsal recorded.

Source: README.md, section "What was not done and limitations"